deficiency during pregnancy can result in neural tube defects like spina bifida and anencephaly.

(f) Cyanocobalamin (B12): The vitamin is named as cyanocobalamin because of the presence of cobalt and cyanide in its structure. It can be absorbed in the body only in the presence of Intrinsic Factor (IF).

- **Functions:** Physiological functions and biochemical functions of cyanocobalmin are:
  - Maturation of erythrocytes
  - Synthesis of myelin (white sheath of lipoprotein) that surrounds many nerve fibres.
  - Increase in White Blood Corpuscle (WBC) count and platelet.
  - Stimulation of appetite and general wellbeing of the people.
  - Cures neurological symptoms of pernicious anemia.
- Food sources: Cyanocobalmin is synthesized by bacteria and is found in foods of animal origin. Liver is the richest source of cyanocobalmin. Meat, fish, kidney, brain and eggs are good sources of cyanocobalmin.



- Deficiency: Inability to produce the intrinsic factor which binds cyanocobalmin leads to pernicious anemia. The red blood cells are macrocytic and the count is often less than 2.5 million. Symptoms include soreness and inflammation of the tongue, paresthesia (numbness and tingling) in fingers and toes, demyelination of the white fibres of the spinal cord and in severe cases degeneration of the spinal cord.

  Other B complex vitamins include biotin, pantothenic acid which do have their vital functions as coenzymes in various biochemical functions of the body

(g) Vitamin C (Ascorbic acid): Ascorbic acid is the chemical name of vitamin C which can be synthesized from glucose but humans depend on their diet for vitamin C as they do not have an enzyme gulonolactone oxidase which catalyzes the conversion reaction.

- **Functions:** Functions of vitamin C include:
  - Collagen formation of bone, teeth, cartilage, skin and scar tissue.
  - Formation of dentin layer of tooth
  - Wound healing.
  - Activation of calcitonin, gastrin, oxytocin, thyrotropin, vasopressin.
  - Drug detoxification
  - Regulation of cholesterol, maintenance of the blood vessel structure and antioxidant effects.

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